

Replication Guide for: *The Forsaken Road: Reassessing Living Standards Following the Cuban Revolution and the American Embargo*

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1 Introduction

Our replication files are structured as follows. `full-replication.do` will run all results for the paper, including those in the appendix. The only exception is the prediction intervals, which use the `scpi` package (Cattaneo et al., 2025) and is run separately in Python. Replicating the whole paper may take several of hours, depending on machine processing capacity.

To replicate the whole study, follow these steps:

1. Open the `full-replication.do` do-file.
2. Change the path to where the “Cuba” folder is stored in your local machine (line 25)
3. Assure the required packages are installed (lines 30-)
4. Run `full-replication.do`.

Doing it this way will call the necessary auxiliary do-files and run the entire study in the most efficient way. If one wishes to inspect the point estimates or recreate only a specific figure or table, the results are already stored in the `results` folder:

- `gdp-results.dta`: Estimates used in Figures 1-7
- `embargo-results.dta`: Estimates used in Figures 8-13

Each `Figure.do` or `Table.do` is a self-contained do-file that will call from these previously-stored results to recreate the figure. One can run them directly, provided they change the paths in each file or define it globally first.

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1.1 Note on Required Packages

All packages can be installed directly from Stata using `ssc install [package]`, with two exceptions. The confidence intervals proposed by [Firpo and Possebom \(2018\)](#) requires a program written by the authors, which does not require installation. We have downloaded the function from Vitor Possobom’s website (see [here](#)) and included in the replication package at `code/function_SCM-CS_v09_stata.do`. Since this program requires several matrix operations, we have also written a wrapper (`code/sc-ci.do`) that automates the construction of confidence intervals by cleaning the data to match the requirements and calling Possobom’s program to handle the calculations.

2 Discrepancies and Errors

In our main results, we use the package `allsynth` ([Wiltshire, 2024](#), Version 1.2). While other packages are available (e.g., `synth2` ([Yan and Chen, 2023](#)), `synth_runner` ([Galiani and Quistorff, 2017](#))), we chose `allsynth` because it is the only package available for Stata that automates bias-correction ([Ben-Michael et al., 2021](#)), demeaning ([Doudchenko and Imbens \(2016\)](#)), and normalization. We decided to keep it throughout for consistency, but results are substantively the same, granted the caveats discussed below, should one choose to replicate with alternative packages.

Crucially, replicating synthetic control studies may lead to small discrepancies, for a few reasons. First, because the original `synth` command by Jens Hainmueller “uses a C++ plugin for numerical optimization, the results might differ slightly on different computers.” ([Yan and Chen, 2023](#), p. 623). Because all packages we aware off (`synth2`, `synth_runner`, and `allsynth`) are wrappers for the original `synth` command, they use `synth` for optimization and this problem can arise regardless of the package used. In practice, this amounts to the numerical optimization plugin choosing marginally different donor weights.¹

Additionally, one common error from the `synth` package that may propagate to all of these wrapper packages is `conformability error r(503)`. It may happen within a non-critical command that handles donor unit labels. We recommend editing the `synth.ado` file in advance for a smooth replication, in the following way:²

- 1 Go into the `ado` folder where Stata saves the packages and find the `synth.ado` file (should be in the folder `ado/plus/s`)
- 2 Go to line 672 and replace `mat rowname 'wsolout' = 'colabels'` with `capture mat rowname 'wsolout' = 'colabels'`
- 3 Save the `synth.ado` file and restart Stata.

¹This study was run in a Apple MacBook M2. `synth` (version 0.0.6 - 08/12/2013), `allsynth` (version 1.2 - 11/07/2024). To check the version you have installed, run: `which synth` and `which allsynth`. We are also aware that because the `synth` command was written in 2013, it does contain a plugin supported by Apple Silicon (M-series) chips. For users of such machines, please follow [these instructions](#) in case of errors.

²More details at: <https://www.statalist.org/forums/forum/general-stata-discussion/general/184880-synth-conformability-error>

We are aware of two differences between packages. First, the `synth` command rounds the estimated donor weights internally (i.e., not only for output display) and use these rounded weights for creating the synthetic counterfactual. Importantly, these weights may not sum to exactly one due to rounding error.³ Of special relevance to our case, the `allsynth` package rescale these weights such that they indeed sum to one, which may create slightly different point estimates (`synth2` and `synth_runner` use the original `synth` weights).⁴

The second source of discrepancies are the ways in which these packages calculate p -values. Critically, the `synth` command does not automate inference. Moreover, [Abadie et al. \(2010, 2015\)](#) only formally propose a “joint” p -value; that is, a single p -value for the whole post-treatment period. In their practical implementations, these authors suggest a scaling to account for pre-treatment fit. Thus, one should compare the distribution of ratios of post-treatment to pre-treatment RMSPE. However, these two articles do not provide a specific formula, and it is only in [Abadie \(2021\)](#) that we get a formal definition.⁵ [Abadie \(2021\)](#)’s definition is (essentially) as follows:

$$\text{Standardized } p\text{-value} = \mathbf{P} \left(\frac{\text{RMSPE}_{\text{post}}^j}{\text{RMSPE}_{\text{pre}}^j} \geq \frac{\text{RMSPE}_{\text{post}}^1}{\text{RMSPE}_{\text{pre}}^1} \right), \quad j \in \{1, 2, 3, \dots, J+1\} \quad (1)$$

$J+1$ is defined as the total number of units, with $j=1$ defined as Cuba and $j=2, \dots, J+1$ as the donors. RMSPE is the root mean squared predictor error; the numerator is defined over the post-treatment period and the denominator is defined over the pre-treatment period. `synth2` and `allsynth` implement this formula for the “joint” p -value case.

However, we are interested in obtaining p -values for *each* post-treatment period. First, we follow [Galiani and Quistorff \(2017\)](#) and calculate a standardized p -value (described below) for *each* post-treatment period t , as opposed to a single average. Thus, instead of using the RMSPE for the entire post-treatment period, we simply use the absolute value of the treatment effect in each period ($|\Delta Y_t^j|$ for donors and $|\Delta Y_t^1|$ for Cuba), standardized by pre-treatment fit. Again, each program handles this process slightly different.

[Galiani and Quistorff \(2017\)](#) proposes excluding the treated unit to the pool of comparisons, because this would require random treatment assignment. They note their metric is not a p -value from classical randomization inference, but rather simply the “proportion of control units that have an estimated effect at least as large as that of the treated unit.” ([Galiani and Quistorff, 2017](#), pp. 836-87). To arrive at the formula proposed by [Abadie](#)

³John Wiltshire annotates this issue on line 1527 of the `allsynth.ado` file.

⁴This was precisely our case. When using `synth`, in some specifications our weights sum to 1.01 due to rounding. The differences are naturally trivial. For instance, the synthetic counterfactuals generated by `synth` and `allsynth` in 1972 are \$4694 and \$4689, respectively. For the rescaling procedure, see the `allsynth.ado` file, lines 1527-1585.

⁵Though we note that one could infer from their language: [Abadie et al. \(2010, p. 503\)](#) writes that “If one were to assign the intervention at random in the data, the probability of obtaining a post/pre-Proposition 99 MSPE ratio as large as California is $1/39 = 0.026$ ”; [Abadie et al. \(2015, p.505\)](#) writes “If one were to pick a country at random from the sample, the chances of obtaining a ratio as high as this one would be $1/17 \simeq 0.059$.”

(2021), one would have to add one to both the nominator and denominator (Galiani and Quistorff, 2017, pp. 836,844).

We follow Galiani and Quistorff (2017) in excluding Cuba from the pool of comparisons and interpret our p -value accordingly. Another consideration is whether to include to include the treated unit as a potential donor for the control units in the permutation procedure. While Galiani and Quistorff (2017) explicitly excludes the treated unit from the donor pool, we follow Wiltshire (2024) and include the treated unit, Cuba. Crucially, note that (i) including Cuba in the pool of comparisons, and (ii) including Cuba as a potential *donor* for control units in placebo permutations, are two separate considerations. The first regards an assumption of random assignment into treatment and affects the denominator (J vs. $J + 1$), while the second directly affects the construction of placebo effects ($\Delta Y_t^T \forall j \neq 1$).

We do so intentionally to provide more conservative p -values, to the extent that the Cuba’s large diversion would asymptotically increase the estimated gaps in the placebo exercise. We emphasize here that our results are robust to both approaches – indeed, in nearly all post-treatment years the effect for Cuba the largest, often by far.

To understand these differences more systematically, here we provide specific definitions of the formulas in each package, and contrast them to the formula we ultimately use in this paper. Again, define ΔY_t^1 as the gap between the treated unit and its synthetic control, ΔY_t^j the analogous measure for each donor $j \in \{2, 3, \dots, J + 1\}$, and $\mathbf{1}(\cdot)$ as an indicator function that returns 1 if true, and 0 if not:

- **synth2** calculates as the number of treatment effects in period t that are of equal or greater magnitude to ΔY_t^1 , *without* scaling for pre-treatment RMPSE:

$$p\text{-value}_t^{\text{synth2}} = \frac{1}{J + 1} \sum_{j=1}^{J+1} \mathbf{1}(|\Delta Y_t^j| \geq |\Delta Y_t^1|) \quad (2)$$

- **allsynth** does similarly, but scales both gaps by the pre-treatment RMSPE (thus comparing the *ratios*), while including the treated unit both in the pool of comparisons ($J + 1$) and as a potential donor in the construction of placebo effects $\Delta Y_t^j \forall j \neq 1$:

$$p\text{-value}_t^{\text{allsynth}} = \frac{1}{J + 1} \sum_{j=1}^{J+1} \mathbf{1}\left(\frac{|\Delta Y_t^j|}{\text{RMSPE}_{\text{pre}}^j} \geq \frac{|\Delta Y_t^1|}{\text{RMSPE}_{\text{pre}}^T}\right), j \in \{1, 2, \dots, J + 1\} \quad (3)$$

with Cuba included as donor in ΔY_t^j

- **synth_runner** offers both unscaled and scaled (which they called “standardized”), but explicitly *excludes* the treatment unit from the pool of comparison and the construction of $\Delta Y_t^j \forall j \neq 1$:

$$p\text{-value}_t^{\text{synth_runner}} = \frac{1}{J} \sum_{j=2}^{J+1} \mathbf{1}(|\Delta Y_t^j| \geq |\Delta Y_t^1|), \quad j \in \{2, 3, \dots, J + 1\} \quad (4)$$

with Cuba excluded as donor in ΔY_t^j

$$\text{Std. } p\text{-value}_t^{\text{synth-runner}} = \frac{1}{J} \sum_{j=2}^{J+1} \mathbf{1} \left(\frac{|\Delta Y_t^j|}{\text{RMSPE}_{\text{pre}}^j} \geq \frac{|\Delta Y_t^1|}{\text{RMSPE}_{\text{pre}}^T} \right), j \in \{2, 3, \dots, J+1\} \quad (5)$$

with Cuba excluded as donor in ΔY_t^j

Since our results use the `allsynth` package, we use its built-in permutation routine – option `pvalues(rmspe)` –, which *does* include Cuba as a potential donor in the calculation of ΔY_t^j . However, we adjust the final calculation to exclude Cuba from the pool of comparisons, following [Galiani and Quistorff \(2017\)](#). Specifically, we use the p -values stored as `p` and subtract 1 from both the nominator and denominator, and save them as `pval`. Thus, our final formula is:

$$\text{Std. } p\text{-value}_t = \frac{1}{J} \sum_{j=2}^{J+1} \mathbf{1} \left(\frac{|\Delta Y_t^j|}{\text{RMSPE}_{\text{pre}}^j} \geq \frac{|\Delta Y_t^1|}{\text{RMSPE}_{\text{pre}}^T} \right) \quad j \in \{2, 3, \dots, J+1\} \quad (6)$$

with Cuba *included* as donor in ΔY_t^j

3 Variables Sources and Definitions

Table 1: Variable Definitions and Sources: `master.dta`

Variable Name	Description
<i>Identifiers and Classification</i>	
<code>countrycode</code>	ISO Country Code
<code>country</code>	Country Name
<code>country_moxlad</code>	Country Name (MOxLAD Database)
<code>id</code>	Unique Country Identifier
<code>latam</code>	Dummy: Latin American Country
<code>former_ussr</code>	Dummy: Former USSR Country
<code>communist</code>	Dummy: Communist/Socialist State
<code>year</code>	Year
<i>GDP, Population, and Maddison Adjustments</i>	
<code>gdppc23</code>	Unadjusted GDP per capita (MPD 2023)
<code>pop23</code>	Population (in thousand, MPD 2023)
<code>adj_mad23</code>	Adj. GDP per capita (MPD 2023)
<code>adj_transf23</code>	Adj. GDP per capita minus Soviet aid (MPD 2023)
<code>gdppc13</code>	Unadjusted GDP per capita (MPD 2013)
<code>adj_mad13</code>	Adj. GDP per capita (MPD 2020)
<code>adj_transf13</code>	Adj. GDP per capita minus Soviet aid (MPD 2013)
<code>devereux_cuba</code>	Cuba GDP Index (1957=100) - Devereux (2021)
<code>transfers</code>	Soviet Aid (as % GDP) - Devereux (2020)
<code>devereux_pr</code>	Puerto Rico GDP Index (1957=100) - Devereux (2019)
<i>Predictor Variables</i>	
<code>schooling</code>	Avg. Years of Schooling - van Leeuwen and van Leeuwen-Li (2014)
<code>urban</code>	Urbanization Rate - UN <i>World Urbanization Prospects</i> (2018)

Continued on next page

Table 1 – continued from previous page

Variable Name	Description
Trade: Exports and Imports (World), in million USD	
exports	Total Exports (in million 1957 USD) - MOxLAD
imports	Total Imports (in million 1957 USD) - MOxLAD
trade	Total Trade with the World (X+M) - MOxLAD
Volumes and Unit Value Indices	
volx	Export Volume (1957=100) - MOxLAD
univx	Unit Value Index: Total Exports (1957=100) - MOxLAD
volm	Import Volume (1957=100) - MOxLAD
univm	Unit Value Index: Total Imports (1957=100) - MOxLAD
Trade: Exports and Imports (World), in million GBP	
exports_th	Total Exports (in million 1957 GBP) - TRADHIST
imports_th	Total Imports (in million 1957 GBP) - TRADHIST
trade_th	Total Trade with the World (X+M) - TRADHIST
US Trade Data, Current Prices (1938-1961)¹	
us_export_sa	US Exports to Country - <i>Statistical Abstract</i>
us_import_sa	US Imports from Country - <i>Statistical Abstract</i>
us_trade_sa	Total US Trade (X+M) - <i>Statistical Abstract</i>
US Trade Data, Current Prices (1962-1972)¹	
us_x_comtrade	US Exports to Country - UN Comtrade
us_m_comtrade	US Imports from Country - UN Comtrade
us_trade_comtrade	Total US Trade - UN Comtrade
US Trade (1957 USD)²	
us_exports	US Exports to Country (1957 Prices)
us_imports	US Imports from Country (1957 Prices)
us_trade	Total US Trade (X+M, 1957 Prices)
US Trade (1957 GBP)	
us_exports_th	US Exports to Country, in million 1957 GBP - TRADHIST
us_imports_th	US Imports from Country, in million 1957 GBP - TRADHIST
us_trade_th	Total Trade with US, in million 1957 GBP - TRADHIST
Deflators, Exchange Rates, and Tariffs	
usdeflator	US Price Deflator - BEA
dollars_per_gbp	USD/GBP Exchange Rate - TRADHIST
gbp_price_index	GBP Deflator (1957=100) - ClioInfra Database

Notes: Sources: MPD is the *Maddison Project Database*. MPD 2023 = Bolt and van Zanden (2024); MPD 2020 = Bolt and Van Zanden (2020); MPD 2013 = Bolt and Van Zanden (2013); MOxLAD = Montevideo-Oxford Latin America Database; UN = United Nations; TRADHIST = Fouquin and Hugot (2016); BEA = U.S. Bureau of Economic Analysis. ¹ This data is coded from the US perspective: exports (US is the origin), imports (US is the destination). ² These variables use data from the *Statistical Abstract of the United States* (1938-1961) and UN Comtrade database (1962-1972).

3.1 GDP and Adjustments

We use 1957 GDP per-capita levels of the various editions of the Maddison Project Database (MPD) adjust onward using the implied growth rates of the index from Devreux (2021). Our adjustment for each year, given in equation form, is as follows:

$$\text{GDP per-capita}_t = \text{GDP per-capita}_{1957} \times \frac{\text{GDP Index}_t}{100} \quad \text{for } t > 1957 \quad (7)$$

See the main text of the paper and Appendix A for historiographical details. For further details, refer to [Devereux \(2021\)](#) and the online supplemental materials ([Devereux, 2020](#)).

`devereux_cuba`: Cuba GDP per capita index (1957=100), from [Devereux \(2021, p. ?\)](#)

`transfers`: Estimates of Soviet and Eastern Bloc Aid (as % GDP), sourced from [Devereux \(2020, p. ?\)](#)

`devereux_pr`: Puerto Rico GDP per capita index, from [Devereux \(2019, p. ?\)](#), rebased to 1957=100.

3.2 US Trade Data

All variables starting with `us_` are coded from the US side, as US Exports to [Destination Country] and US Imports from [Origin Country]. Thus, the latter is used in the main results as Cuban Exports to the US. Because data from the UN Comtrade Database is only available starting in 1962, we supplement it (1938-1961) using data from the *Statistical Abstract of the United States*. Both datasets technically cover trade with the United States and Puerto Rico, until 1980. Also see 3.4, below.

UN Comtrade Database: Data is available at <https://comtradeplus.un.org/TradeFlow>. We performed the query as shown below in Figure 1. The resulting data is saved, unedited, at `data/UNComtrade 1962-1973.xlsx`

The screenshot shows the UN Comtrade Database query interface. The page has a dark header with the UN Comtrade Database logo and navigation links. The main content area is titled 'Trade Data' and includes a warning message: 'Caution: The results depend on available reported data, and the level of detail may vary.' Below this, there is an 'Information' section with a dropdown menu. The 'What are you looking for?' section includes filters for 'Type of Product' (Goods, Services), 'Frequency' (Annual, Quarterly), and 'Classification' (HS, SITC, BIC, SITC LTR). The 'Refine your search' section includes filters for 'SITC (as reported) Commodity Codes' (12024), 'Periods (year, month)' (1962, 1963, 1964, 1965, 1966, 1967, 1968, 1969, 1970, 1971, 1972, 1973, 1974, 1975), 'Partners' (US), 'Trade Flows' (Exports, Imports), 'Modes of Transport' (Sea/Air/Sea/Air/Sea/Air), and 'Customs Codes' (SITC/Customs codes). The 'Set your query options' section includes filters for 'Breakdown trade' (Plus) and 'Aggregate By' (None). At the bottom, there are buttons for 'Preview', 'Download', 'Delivery', and 'Reset'.

Figure 1: Query in UN Comtrade Database

Statistical Abstract of the United States: Data is available at https://www.census.gov/library/publications/time-series/statistical_abstracts.html. The consulted chapters have been downloaded and are available in the `/data/US Statistical`

Abstract/ folder. Exact tables and page numbers are provided below, with filenames for each yearbook indicated in parenthesis.

`us_export_sa`: US Exports to [Destination]

- **1938-1942**: Table 614, p. 550, 1944-1945 Yearbook (1944-08.pdf).
- **1943-1947**: Table 977, pp. 872-73, 1949 Yearbook (1949-12.pdf);
- **1948-1955**: Table 1132, pp. 914-915, 1956 Yearbook (1956-13.pdf);
- **1956-1959**: Table 1194, pp.896-897, 1960 Yearbook (1960-11.pdf);
- **1960-1963**: Table 1229, pp. 876-877, 1962 Yearbook (1964-13.pdf).

`us_import_sa`: US Imports from [Origin]

- **1938-1942**: Table 615, p. 554, 1944-1945 Yearbook (1944-08.pdf);
- **1943-1947**: Table 977, pp. 872-73, 1949 Yearbook (1949-12.pdf);
- **1948-1955**: Table 1132, pp. 914-915, 1956 Yearbook (1956-13.pdf);
- **1956-1959**: Table 1194, pp.896-897, 1960 Yearbook (1960-11.pdf);
- **1960-1963**: Table 1229, pp. 876-877, 1962 Yearbook (1964-13.pdf).

3.3 Trade with the World Data

Data on Cuba's trade with the world comes from the Montevideo-Oxford Latin America Database (Rey and Bértola, 2018), available at <https://bancodedatos-fcs.shinyapps.io/MOXLAD/>.

`exports` and `imports` are the variables “Exportaciones (millones de USD corrientes)” and “Importaciones (millones de USD corrientes)”, respectively. The original data is measured in million of current USD, and we deflated to million 1957 USD. `trade` was created as `exports + imports`, again measured in million 1957 USD.

`univx` and `unitvm` are unit value indexes of exports and imports, respectively, downloaded as “Índice de valor unitario de exportación (1970=100)” and “Índice de valor unitario de importación (1970=100)” and rebased to 1957.

`volx` and `volm` are indices of exported/imported volume, downloaded from “Índice de volumen exportado (1970=100)” and “Índice de volumen importado (1970=100)”, respectively, and rebased to 1957.

3.4 Alternative Trade Data

Our second source of trade flows is the TRADHIST database (Fouquin and Hugot, 2016). It contains 1.9 million bilateral trade follows from 1827-2014. Importantly, this data is measured in pounds sterling. We deflated to 1957 prices using British Pound price index from ClioInfra (see below). The deflator is available at `data/gbp-deflator.dta`.

The underlying sources in TRADHIST are the following: US-Cuba bilateral flows come from the Carter et al. (2006) (1827-1970), and for 1971 onward from the International Monetary Fund Direction of Trade Statistics (DOTS). Total Cuban imports are sourced from Mitchell (2007) (1900-59) and then from World Bank's World Development Indicators. Exports are sourced again from Mitchell (2007) (1900-47), IMF DOTS (1948-1959), and World Bank's WDI (1960 onward).

Data for 1939 is missing for Venezuela and El Salvador. To avoid losing these two countries as donors, we interpolate the data for these two observations. We use Stata's `ipolate` command, which creates a linear projection based on a reference variable – in this case, we used `us_trade` and `trade`.

3.5 Deflators, Exchange Rates, and Tariffs

`usdeflator`: U.S. Bureau of Economic Analysis, Gross national product (implicit price deflator) [A001RD3A086NBEA], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/A001RD3A086NBEA>.

`dollars_per_gbp`: `XCH_RATE_d` from TRADHIST (Fouquin and Hugot, 2016).

`gbp_price_index`: GBP price index, from the Clio-Infra database, available at <https://clio-infra.eu/Indicators/Inflation.html#>. Re-based to 1957=100.

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